

13.11.25

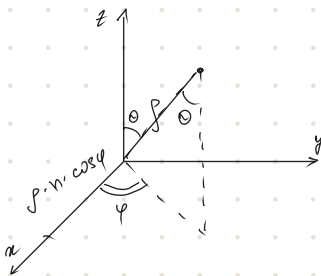
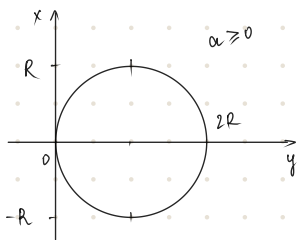
Пр. 9

Задачи самостоятельно разобрать

Крив. инт. 1го рода  
составляем конспект.~3550  $\Omega$  - часть шара  $x^2 + y^2 \leq R^2$ , или внутри цилиндра  
( $x^2 + y^2$ )<sup>2</sup> =  $R(x^2 + y^2)$  ( $x \geq 0$ )

$$\iiint_{\Omega} f \, dv = \iiint f(\underbrace{\rho \cos \varphi \sin \Theta}_{x}; \underbrace{\rho \sin \varphi \sin \Theta}_{y}; \underbrace{\rho \cos \Theta}_{z}) d\rho d\varphi d\Theta \leftarrow \text{сферич. коорд.}$$

$$\rho^2 \sin \Theta \, d\rho \, d\varphi \, d\Theta$$



$$\begin{aligned} -\frac{\pi}{2} &\leq \varphi \leq \frac{\pi}{2} \\ 0 &\leq \rho \leq R \\ -\frac{\pi}{2} &\leq \Theta \leq \frac{\pi}{2} \end{aligned}$$

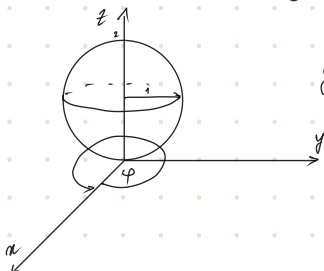
$$\equiv \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} d\varphi \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sin \Theta \, d\Theta \int_0^R f \rho^2 \, d\rho$$

~3551  $\Omega$  - две. часть 2х шаров  $x^2 + y^2 \leq R^2$  и  $x^2 + y^2 + (z - R)^2 \leq R^2$ 

$$0 \leq \Theta \leq \frac{\pi}{2}$$

$$0 \leq \rho \leq \frac{\sqrt{3}R}{2}$$

$$-\frac{\pi}{2} \leq \varphi \leq \frac{\pi}{2}$$

для случая, когда  $R=2: z=1$ .  $x^2 + y^2 = 3$ ;  $R=\sqrt{3}$   
 $0 \leq \varphi \leq \pi$ 

$$\begin{cases} x^2 + y^2 + (z - R)^2 = R^2 \\ x^2 + y^2 + z^2 = R^2 \end{cases} \quad (*)$$

$$\begin{aligned} 2Rz &= R^2 \\ z &= R/2 \end{aligned}$$

$$\text{В м-м} \quad z = \frac{R}{2}$$

$$(*) \quad x^2 + y^2 = \frac{3R^2}{4}$$

$$\ominus \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} d\varphi \int_0^{2\pi} \sin\Theta d\Theta \int_0^{\sqrt{3}R} f\rho^2 d\rho$$

$$x^2 + y^2 + z^2 = R^2$$

$$\begin{cases} x = \rho \sin\Theta \cos\varphi \\ y = \rho \sin\Theta \sin\varphi \\ z = \rho \cos\Theta \end{cases}$$

$$\begin{aligned} \rho^2 (\sin^2\Theta + \cos^2\Theta) &= R^2 \\ \rho^2 \sin^2\Theta \cos^2\varphi + \rho^2 \sin^2\Theta \sin^2\varphi + \rho^2 \cos^2\Theta &= \\ &= \rho^2 \sin^2\Theta + \rho^2 \cos^2\Theta = \rho^2 \end{aligned}$$

$$\begin{aligned} \rho^2 \sin^2\varphi + (\rho \cos\Theta - R)^2 &= R^2 \\ \rho^2 \sin^2\varphi + \rho^2 \cos^2\Theta - 2R\rho \cos\Theta + R^2 &= R^2 \end{aligned}$$

$$\rho^2 = \frac{2R \cos\Theta}{\sin^2\varphi + \cos^2\Theta}$$

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$$\iiint_{\Omega} (x^2 + y^2) dx dy dz = \int_0^{2\pi} d\varphi \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sin\Theta d\Theta \int_0^R \rho^2 \sin^2\Theta \rho^2 d\rho, \quad \Omega - \text{верхн. половина шара}$$

$$x^2 + y^2 + z^2 = R^2$$

$$0 \leq \rho \leq R$$

$$0 \leq \Theta \leq \frac{\pi}{2}$$

$$0 \leq \varphi \leq 2\pi$$

$$\begin{cases} x = \rho \sin\Theta \cos\varphi \\ y = \rho \sin\Theta \sin\varphi \\ z = \rho \cos\Theta \end{cases} \quad \begin{array}{l} \varphi - \text{угл. перем.} \\ \kappa - \text{сферич.} \\ \text{коорд.} \end{array}$$

$$x^2 + y^2 = \rho^2 \sin^2\Theta$$

$$\begin{aligned} \ominus \int_0^{2\pi} d\varphi \int_0^{\frac{\pi}{2}} \sin^2\Theta d\Theta \int_0^R \rho^4 d\rho &= 2\pi \frac{R^5}{5} \int_0^{\frac{\pi}{2}} \sin^3\Theta d\Theta = 2\pi \frac{R^5}{5} \int_{\frac{\pi}{2}}^0 \sin\Theta d(\cos\Theta) = \\ &= 2\pi \frac{R^5}{5} \left( \cos\Theta - \frac{\cos^3\Theta}{3} \right) \Big|_0^{-\frac{\pi}{2}} = 2\pi \frac{R^5}{5} \left( 1 - \frac{1}{3} \right) = 2\pi \frac{R^5}{5} \cdot \frac{2}{3} = \frac{4\pi R^5}{15} \end{aligned}$$

Криволинейн. интеграл по дуге

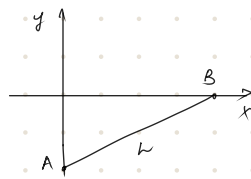
~ 3770  $L$  - вып. прямой  $y = \frac{1}{2}x - 2$ , заключ. между  $M, A(0; -2)$  и  $B(4; 0)$

$$\int_L \frac{dx}{x-y} = \frac{\sqrt{5}}{2} \int_0^4 \frac{1}{x - (\frac{1}{2}x - 2)} dx = \frac{\sqrt{5}}{2} \int_0^4 \frac{1}{\frac{x}{2} + 2} dx = \frac{\sqrt{5}}{2} \int_0^4 \frac{2}{x+4} dx = \left\{ \right.$$

$$ds = \sqrt{1 + \left(\frac{1}{2}\right)^2} dx = \sqrt{1 + \frac{1}{4}} dx = \frac{\sqrt{5}}{2} dx$$

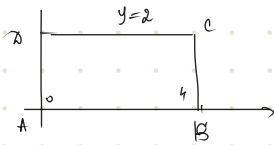
$$y' = \frac{1}{2}$$

$$\int_0^4 \sqrt{5} \ln |x+4| dx = \sqrt{5} \ln 2$$



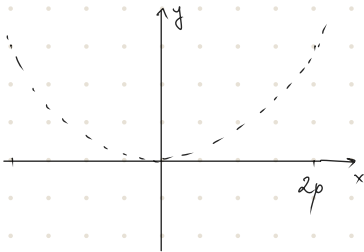
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$\int_L xy \, ds$ ,  $L$  - контур прямоуго. в верш.  $A(0,0), B(4,0), C(4,2), D(0,2)$



$$\begin{aligned} \int_L xy \, ds &= \int_{AB} + \int_{BC} + \int_{CD} + \int_{DA} = \int_0^4 x \cdot 0 \cdot \sqrt{1+0^2} \, dx + \\ &+ \int_0^2 4 \cdot y \, dy + \int_4^0 x \cdot 2 \cdot \sqrt{1+0^2} \, dx + \int_0^2 0 \cdot y \, dy = \\ &= x^2 \Big|_4^0 = -16 \end{aligned}$$

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$$\begin{aligned} x^2 &= 2py \\ y^2 &= 2px \end{aligned}$$

$$\begin{aligned} 2p \sqrt{2px} &= x^2 \\ 4p^2 2p x &= x^4 \\ 8p^3 &= x^3 \\ x=0 &\quad \text{или} \quad x=2p \end{aligned}$$

$$y = \sqrt{2px} \quad y' = \frac{p}{\sqrt{2px}}$$

$$\begin{aligned} \int_L y \, ds &= \int_0^{2p} \sqrt{2px} \sqrt{1 + \frac{p^2}{2px}} \, dx = \int_0^{2p} \sqrt{2px+p^2} \, dx = \sqrt{p} \int_0^{2p} \sqrt{2x+p} \, dx = \\ &= \frac{1}{2} \sqrt{p} \int_0^{2p} \sqrt{2x+p} \, d(2x+p) = \frac{1}{2} \sqrt{p} \left. \frac{\sqrt{(2x+p)^3}}{3/2} \right|_0^{2p} = \sqrt{p} \left( \frac{\sqrt{(2 \cdot 2p+p)^3}}{3} \right) = \\ &= \frac{(2 \cdot 0 + p)^3}{3} \end{aligned}$$